

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

Nov-Dec, 2013

EE101 Fundamentals of Electrical & Electronics Engineering

Date: 05/12/2013

Time: 10:00 a.m. To 01:00 p.m.

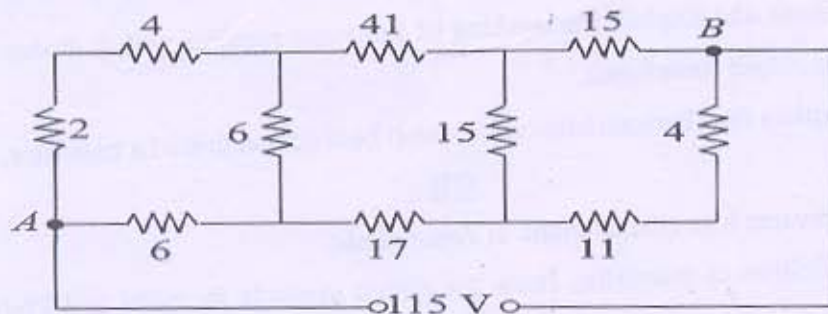
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the factors affecting the resistance of a Conducting material. [04]
- (b) Determine the current in 17 ohm Resistor in the network shown in figure. (All resistance values are in ohms.) [03]



- Q - 2 (a) Explain the charging and discharging of a capacitor, C, through a resistor R, with neat sketch and derive the equation $V_c = V (1 - e^{-t/RC})$. Assume that the R-C series circuit is connected across a d.c. supply of voltage V. [07]
- (b) A parallel plate capacitor has plates of area 2 m^2 spaced by three slabs of different materials. The relative permittivity are 2, 3, and 6 and thicknesses are 0.4, 0.6, and 1.2mm respectively. Calculate the equivalent capacitance and electric stress in each material when the applied voltage is 1000V. [07]

OR

- Q - 2 (a) Explain the following terms: [07]
- i) Magnetic Force
 - ii) Magnetic field Intensity
 - iii) Reluctance
 - iv) Permeability
- (b) Compare Electric and Magnetic Circuit with similarities and dissimilarities. [07]
- Q - 3 (a) Draw the B-H curve (Magnetizing curve) of a magnetic material and label the figure. [07]
- Define all the terms relevant to this curve.

- (b) Define the following terms with respect to a a.c. waveform. [07]
(i) Frequency (ii) Power factor (iii) R.M.S. value
(iv) Amplitude (v) Average Value (vi) Form Factor (vii) Q-factor.

OR

- Q - 3 (a) Explain the following Laws. (any 4) [07]
(i) KCL (ii) KVL (iii) Coulomb's 1st Law (iv) Coulomb's 2nd Law
(v) Faraday's Law (v) Fleming's Left Hand Rule
(b) Prove that the purely capacitive Circuit power consumed is zero when a.c. voltage is applied. Draw relevant phasor diagram and waveforms. [07]

SECTION - II

- Q - 4 (a) Explain how N type semiconductor is formed. Show how it is represented. State its majority and minority charge carriers. [04]
(b) Explain the effect of forward bias and reverse bias on the depletion region of PN junction diode. [03]
Q - 5 (a) Draw the circuit and Explain the working of full wave rectifier with 2 diodes. Draw the input and output waveforms. [07]
(b) Draw and explain the characteristics of common base connection of a transistor. [07]

OR

- Q - 5 (a) Explain the reverse bias characteristic of Zener diode. [07]
(b) Give the definition of transistor. Draw the circuit symbols for NPN and PNP type transistor. [03]
(c) Explain the working principle of transistor with neat diagram. [04]
Q - 6 (a) Explain the half wave rectifier with neat diagram [07]
(b) State the difference between analog and digital signals. State the examples of digital system. [07]

OR

- Q - 6 (a) Convert the following number into equivalent binary numbers [07]
1. 96.25
2. 135
(b) Explain with example the binary addition and subtraction. [07]
